Python OOP & Data Structures Lab Guide

Core Principles, Class Architecture, Magic Methods & Exercises

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1. Core Object-Oriented Principles

• Encapsulation: Bundle data (attributes) and behavior (methods) together.

• Inheritance: Derive specialized classes from parent base classes.

• Polymorphism: Treat derived classes uniformly via duck typing or overrides.

• Abstraction: Hide internal implementation using clean public interfaces.

2. Essential Dunder Methods (__init__, __str__, __repr__)

• __init__(self, ...): Constructor initializing instance attributes.

• __repr__(self): Unambiguous representation primarily for developers and debugging.

• __str__(self): Readable, user-friendly representation for printing.

• __eq__(self, other): Custom equality comparison based on attributes.

3. Best Practices for College Lab Exercises

• Always use type hints for clarity: def calculate_grade(scores: list[float]) -> str:

• Prefer composition over deep inheritance hierarchies where feasible.

• Use docstrings (PEP 257) on all public functions and classes.